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Basic Principles of Medicine 1

Module: Musculoskeletal System | Lecture 12

Posterior Abdominal Wall

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Recommended Reading

- Drake, Vogl, Mitchell. Gray's Anatomy for Students. 4th ed. Elsevier Churchill Livingstone. Chapter 4.

- ***Pay special attention to the "In the Clinic" boxes for the clinical correlations for this topic***

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Objectives

SOM.MKII.BPM1.1.MSK.1.ANAT.1263	Describe the boundaries of the posterior abdominal wall.
SOM.MKII.BPM1.1.MSK.1.ANAT.1264	Describe the attachment, innervation and blood supply of the diaphragm.
SOM.MKII.BPM1.1.MSK.1.ANAT.1265	Describe the attachment sites and the peritoneal lining of the diaphragm.
SOM.MKII.BPM1.1.MSK.1.ANAT.1266	List the innervations, blood supply, general attachments and actions of the muscles of the posterior abdominal wall.
SOM.MKII.BPM1.1.MSK.1.ANAT.1267	Describe the origin of the lumbar plexus and list its branches and their distributions.
SOM.MK.II.BPM1.1.MSK.1.ANAT.1507	Describe the sensory and motor arcs of the cremasteric reflex
SOM.MKII.BPM1.1.MSK.1.ANAT.1268	List the parietal and visceral branches of the abdominal aorta and use this information to identify them on cadaveric specimens and imaging studies.
SOM.MKII.BPM1.1.MSK.1.ANAT.1269	List the approximate vertebral levels of the visceral branches of the abdominal aorta and its bifurcation into the common iliac arteries and their distribution.
SOM.MKII.BPM1.1.MSK.1.ANAT.1270	Describe the inferior vena cava and its tributaries and list the vertebral level where the common iliac veins join.
SOM.MKII.BPM1.1.MSK.1.ANAT.1271	Describe the cisterna chili as well as the thoracic duct and use this information to interpret clinical signs and symptoms.
SOM.MKII.BPM1.1.MSK.1.ANAT.1272	Explain how venous blood may reach the right atrium when the IVC is obstructed

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Lecture Overview

- **Abdominopelvic cavity & boundaries**
- **Diaphragm**
  - Attachments, openings (T8, T10, T12)
  - Blood supply & innervation (C3–C5 phrenic nerve)
- **Muscles of the posterior abdominal wall**
- **Lumbar plexus**
  - Origin & major branches (subcostal, iliohypogastric, ilioinguinal, femoral, obturator, genitofemoral, lateral cutaneous, lumbosacral trunk)
  - Structures supplied
  - Cremasteric reflex
- **Abdominal aorta**
  - Parietal & visceral branches, vertebral levels, bifurcation at L4
- **Inferior vena cava**
  - Tributaries & drainage patterns
  - Azygos system & collateral venous pathways in IVC obstruction
- **Lymphatics**
  - Lumbar nodes, cisterna chyli, thoracic duct

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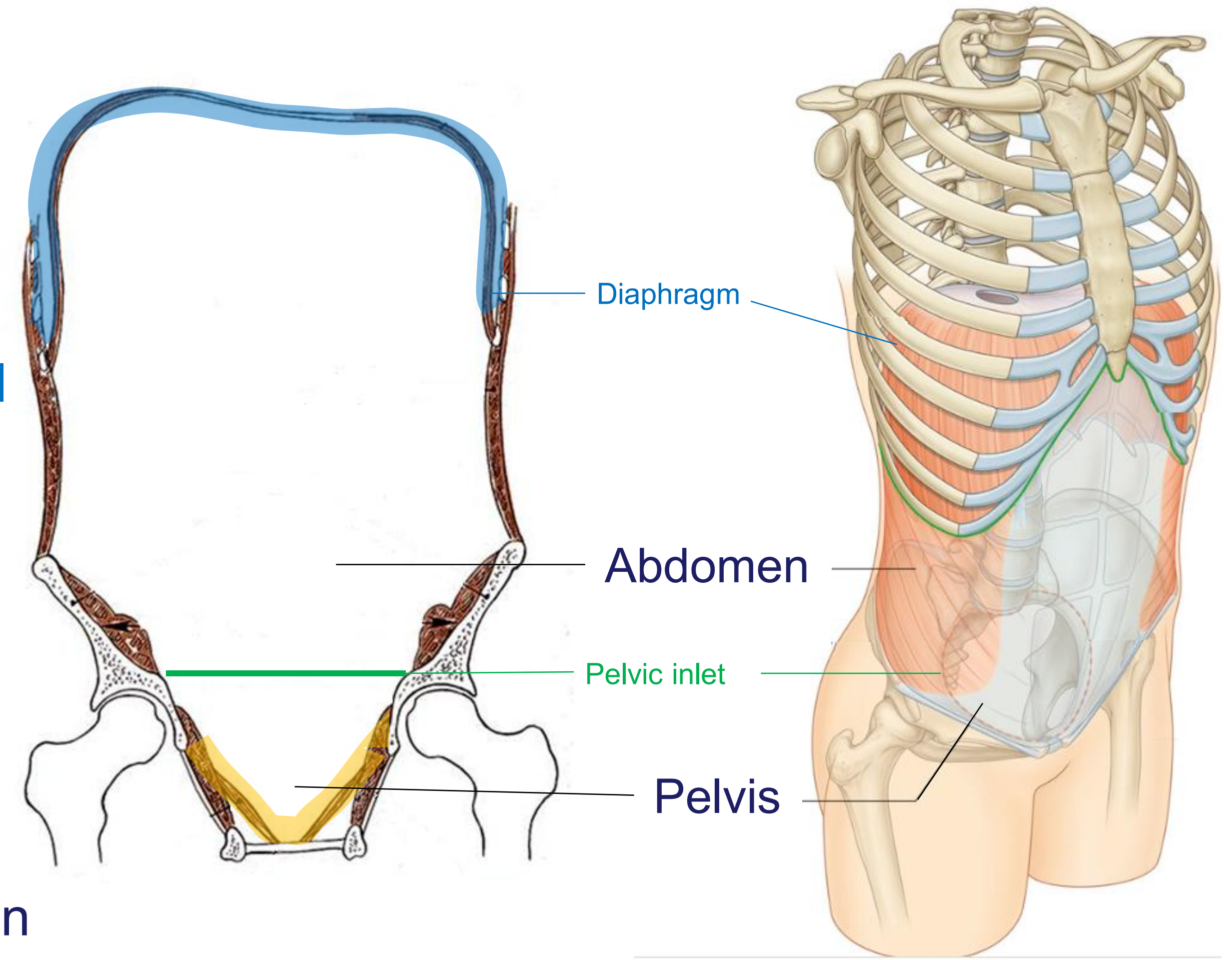
Abdominopelvic cavity

The abdominal and pelvic cavities are continuous



Upper boundary

Thoraco-abdominal diaphragm

Lower boundary

pelvic diaphragm

Pelvic inlet

the division between the abdominal and pelvic cavities

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Diaphragm

Muscular and tendinous separation between the thorax and abdomen

Lined with peritoneum (inferiorly), except for the area superior to the bare area of the liver

Three large openings;

Inferior vena cava	T8 vertebra level
Esophagus	T10 vertebra level
Aorta	T12 vertebra level

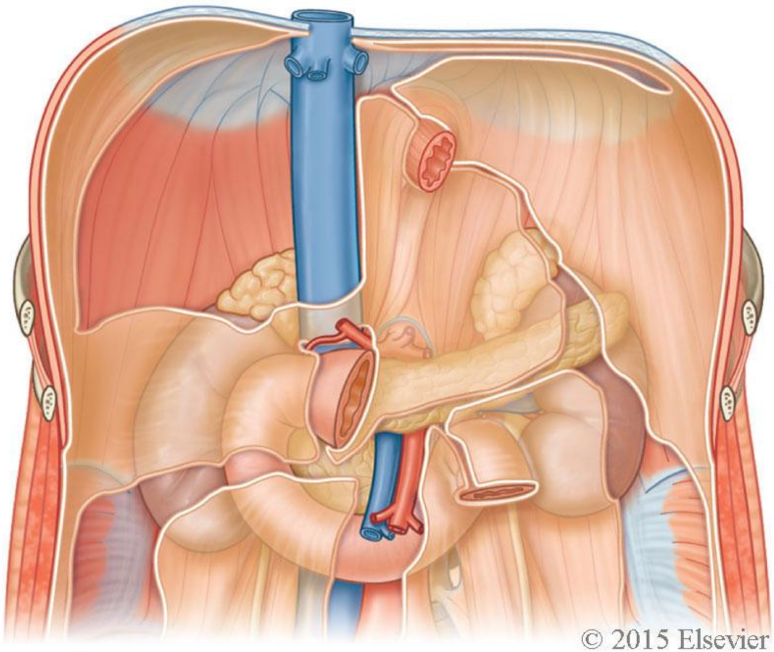
Diaphragm as a muscle, changes the size of the thoracic cavity to facilitate airflow of the lungs during breathing

Blood supply:

- superior and inferior phrenic arteries
- musculophrenic arteries
- pericardiophrenic arteries

Innervation:

- phrenic nerves *(C3, 4, 5)*



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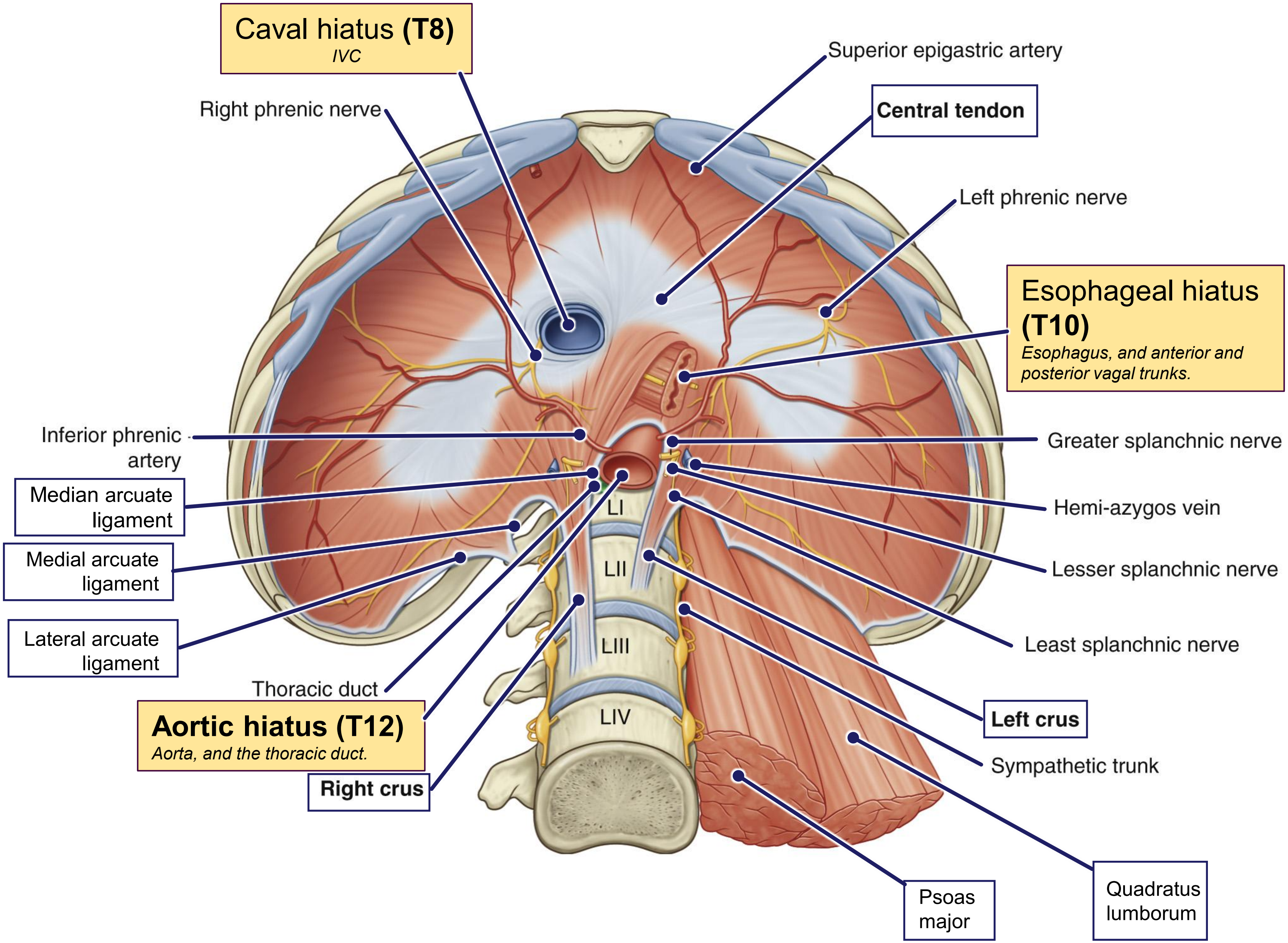

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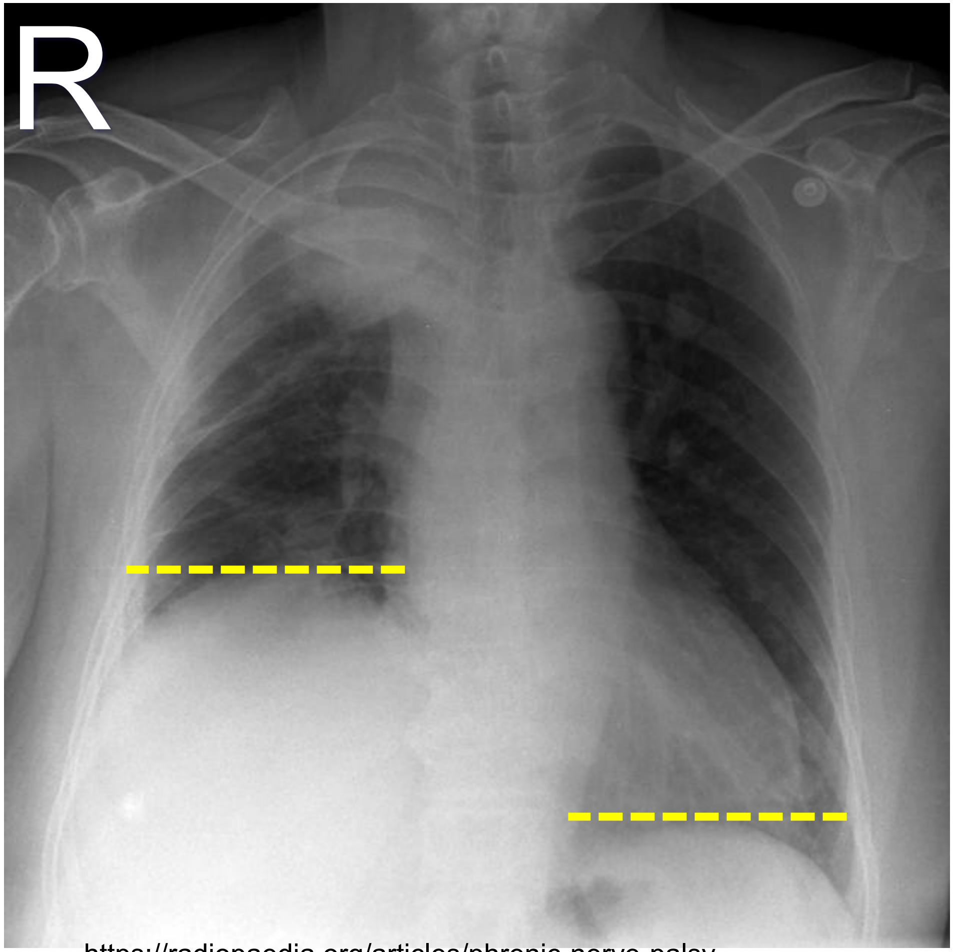
Right phrenic nerve palsy



https://radiopaedia.org/articles/phrenic-nerve-palsy

- Right dome of diaphragm higher than left
- Apical lung mass is likely causing the damage to the phrenic nerve

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Muscles and landmarks of the posterior abdominal wall

Psoas major

Origin = vertebral bodies & transverse processes T12-L5

Insertion = Lesser trochanter

Innervation = ventral rami of L1-3

Action: flexes hip

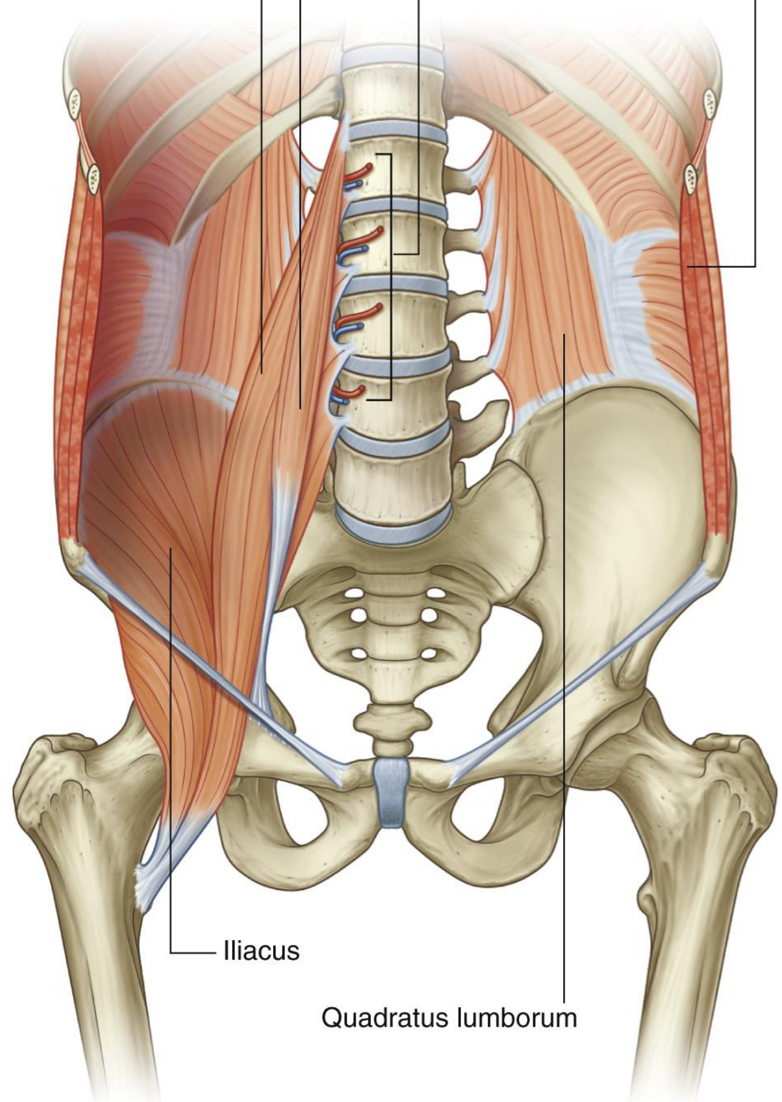
Psoas minor

O = Lateral surface of bodies of T 12 and L1 vertebrae and intervertebral disc

I = Pectineal line and iliopubic eminence

Inn = ventral rami of L1

A = Weak flexion of lumbar vertebral column



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Muscles and landmarks of the posterior abdominal wall

Quadratus lumborum

O = Transverse process of L5, iliac crest and iliolumbar ligament

I = transverse processes of L1-4 and rib 12

Inn = ventral rami of T12-L4

A = Depresses & stabilizes rib 12, lateral bending (*lateral flexion*)

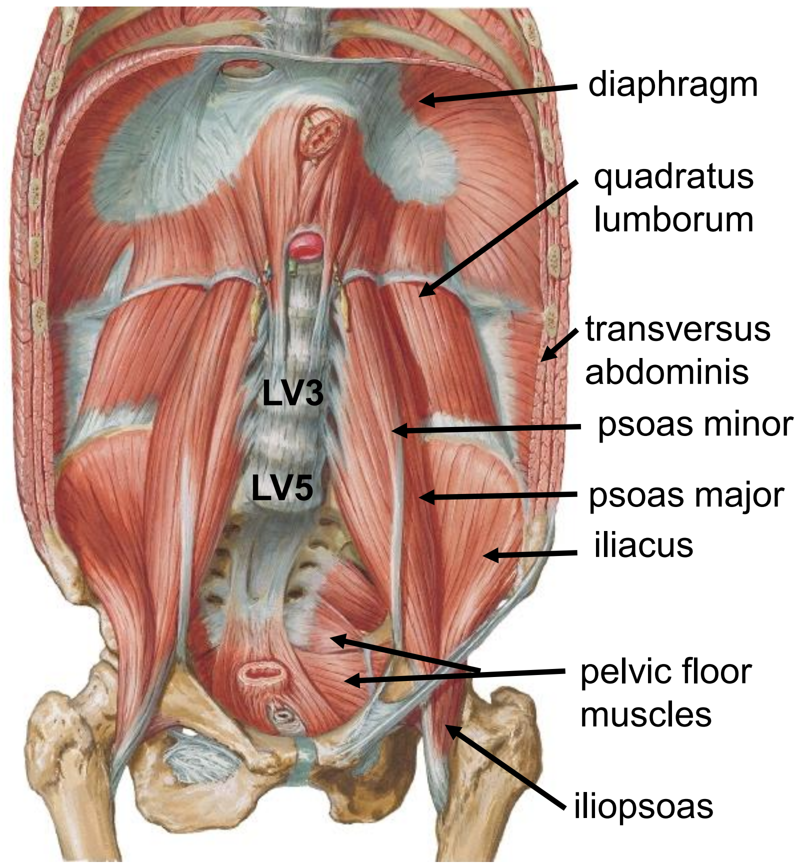
Iliacus

O = Iliac fossa and upper portion of lateral sacrum

I = lesser trochanter

Inn = femoral nerve

Action: flexion of the hip



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Lumbar plexus

- The nerve supply of the posterior abdominal wall, pelvic walls, pelvic floor, and lower limb are closely related.
- The **lumbar plexus** serves the **abdomen**, **anterior**, and **medial thigh**.
- The **sacral plexus** receives a contribution from the lumbar plexus – **lumbosacral trunk**
- The sacral plexus supplies the **pelvic muscles**, **gluteal region**, and **posterior lower limb**

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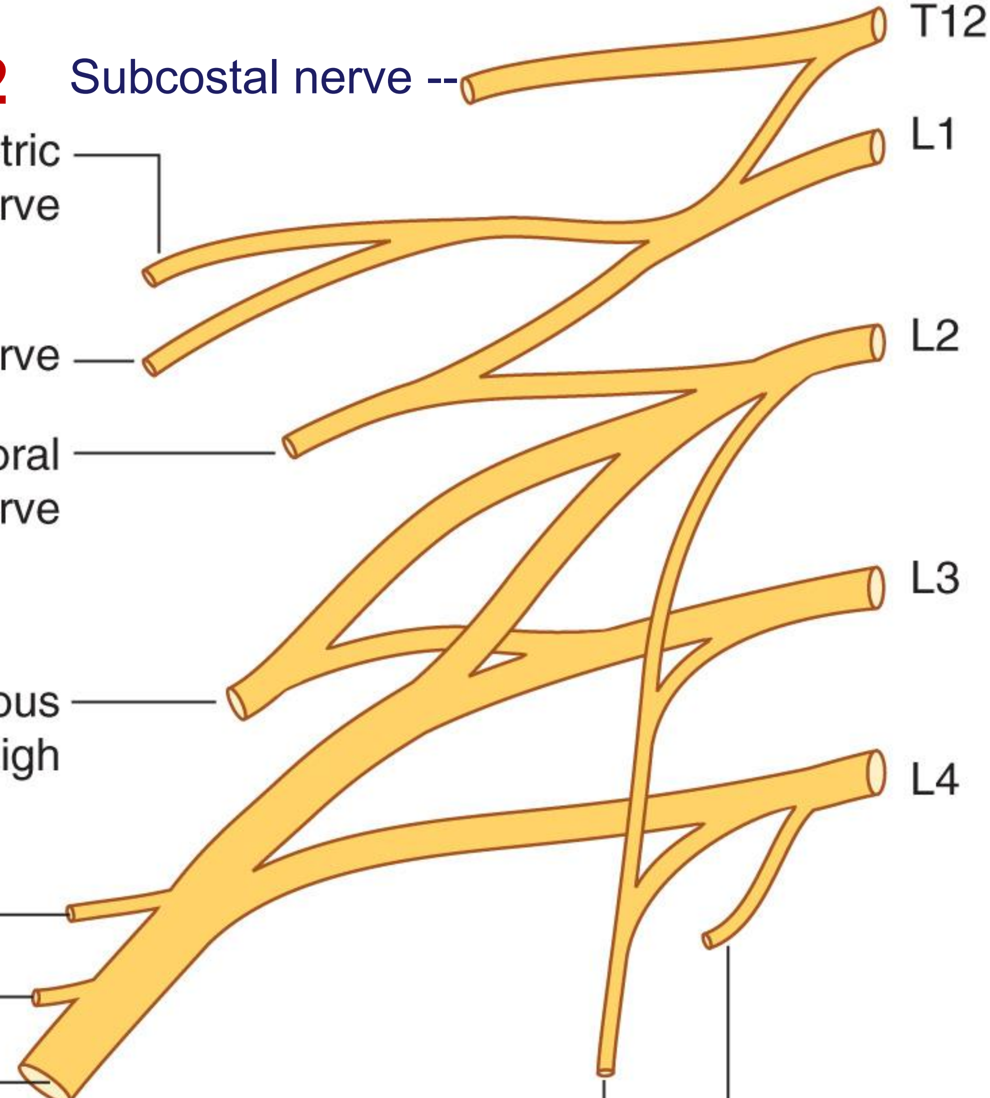
Lumbar plexus



- **T12** Subcostal nerve --
- **T12, L1** Iliohypogastric nerve
- **L1** Ilio-inguinal nerve
- **L1, L2** Genitofemoral nerve
- **L2, L3** Lateral cutaneous nerve of thigh
- To iliacus muscle
- **L2, L3, L4** Femoral nerve
- **L2, L3, L4** Obturator nerve
- **L4, L5** To lumbosacral trunk

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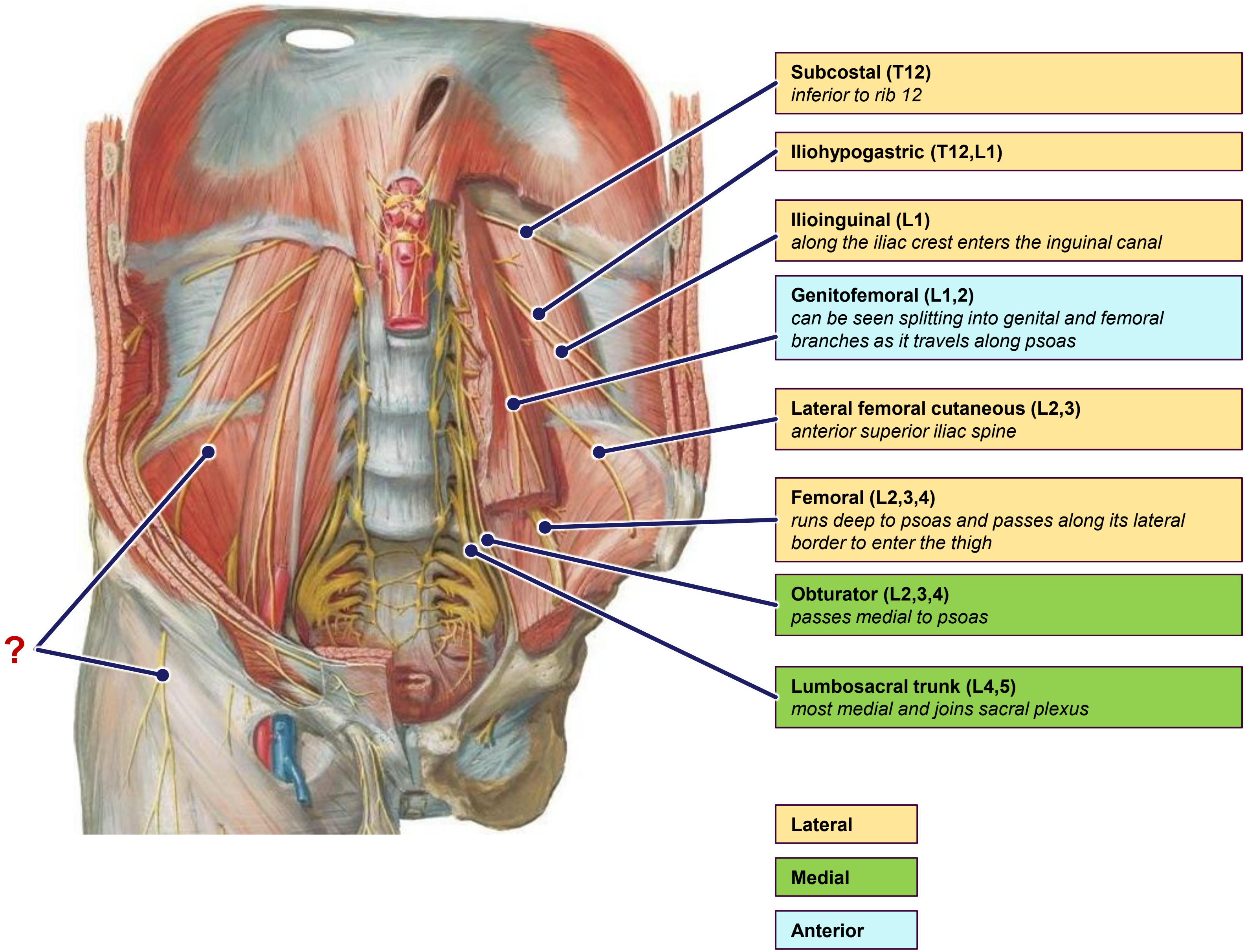

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Lumbar plexus

IN THE LAB

- The femoral nerve *(L2, 3, 4)* may be identified lateral to the psoas major muscle
- The obturator nerve *(L2, 3, 4)* is located medial to psoas major and can better be identified in the pelvis
- The lumbosacral trunk *(L4, 5)* wraps around the sacral ala closely and appears flat
- The iliohypogastric and ilioinguinal nerves *(the L1 nerve)* arise from the same root and may travel some distance before splitting
- The lateral femoral cutaneous nerve *(L2, 3)* is directed towards the ASIS and leaves the pelvis just medial to the ASIS

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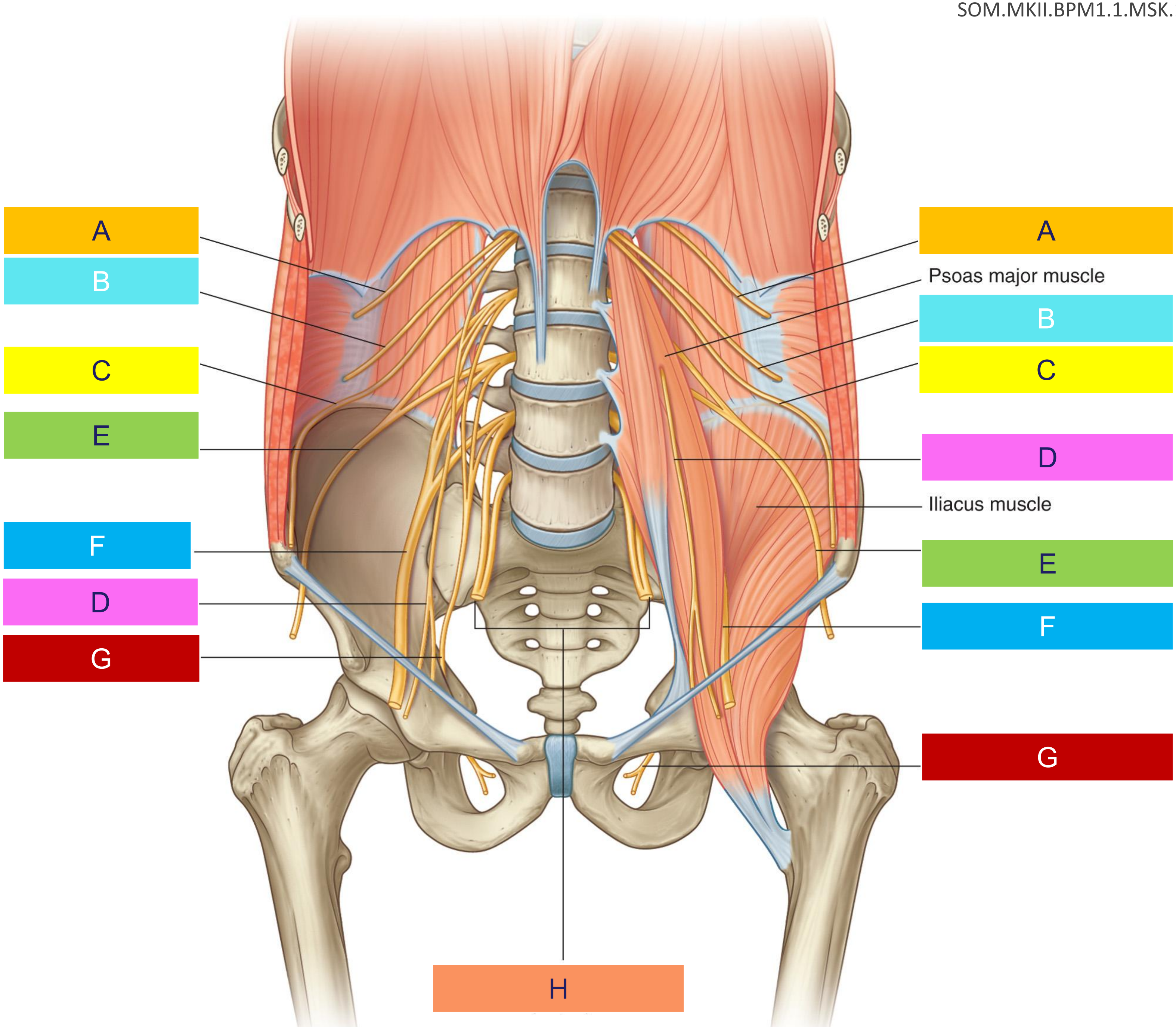

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A	Subcostal
B	Iliohypogastric
C	Ilioinguinal
D	Genitofemoral
E	Lateral femoral cutaneous
F	Femoral
G	Obturator
H	lumbosacral trunk

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Structures supplied by the lumbar plexus

Branch	Motor	Sensory
Iliohypogastric	Internal oblique Transversus abdominis	Posterolateral **gluteal** skin and skin in **pubic** region
Ilio-inguinal	Internal oblique Transversus abdominis	Skin in the **upper medial** thigh, and either the skin over the root of the penis and anterior **scrotum** or the **mons pubis** and **labium majus**
Genitofemoral	Genital branch — cremasteric muscle (in males)	Genital branch—skin of anterior **scrotum** or skin of **mons pubis** and **labium majus**; Femoral branch—skin of **upper anterior** thigh
Lateral cutaneous nerve of thigh		Skin on anterior and lateral thigh to the knee
Obturator	Obturator externus, pectineus, and muscles in **medial compartment of thigh**	Skin on **medial** aspect of the thigh
Femoral	**Iliacus**, pectineus, and muscles in **anterior compartment of thigh**	Skin on **anterior** thigh and medial surface of leg

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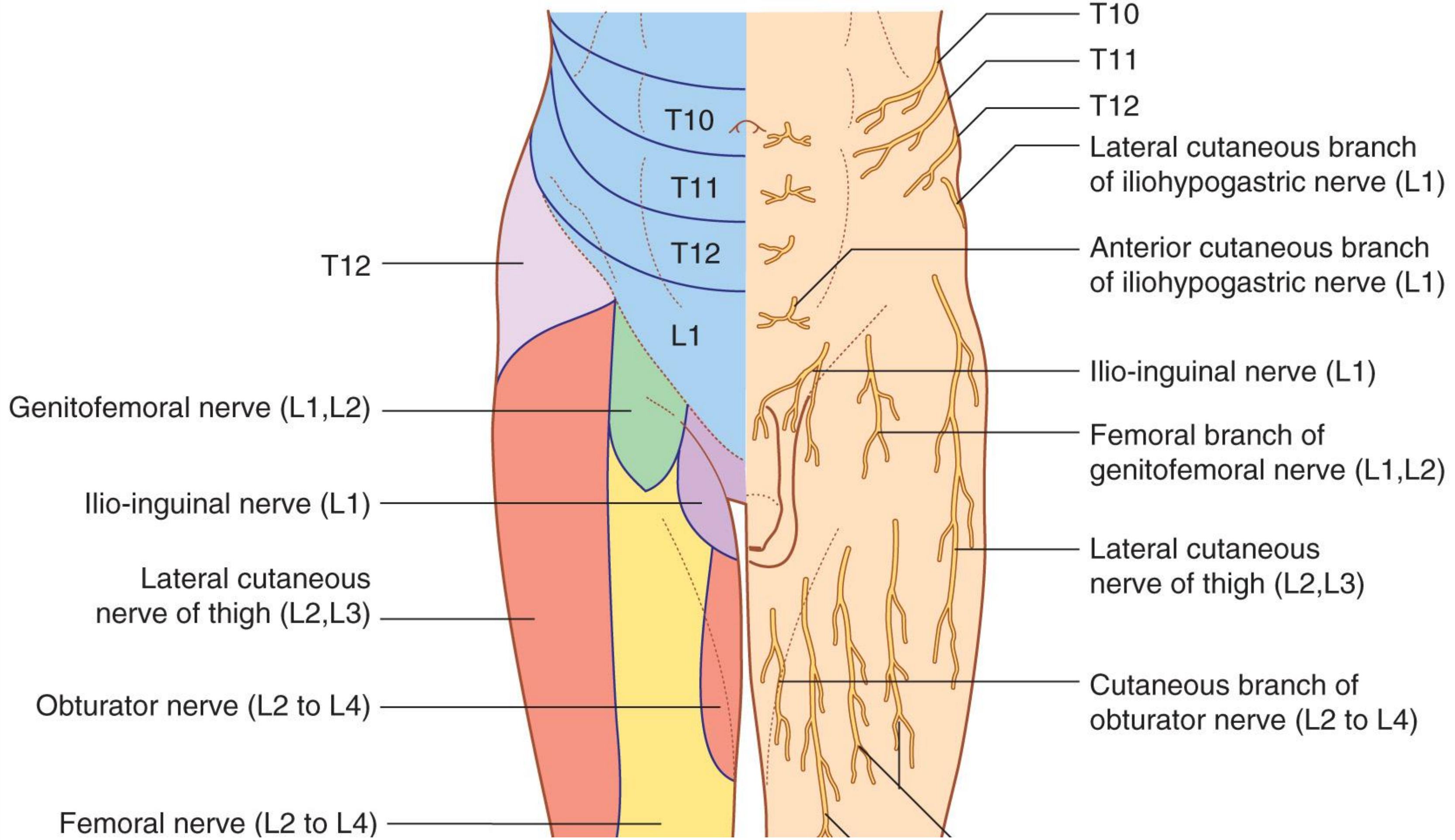
Structures supplied by the lumbar plexus



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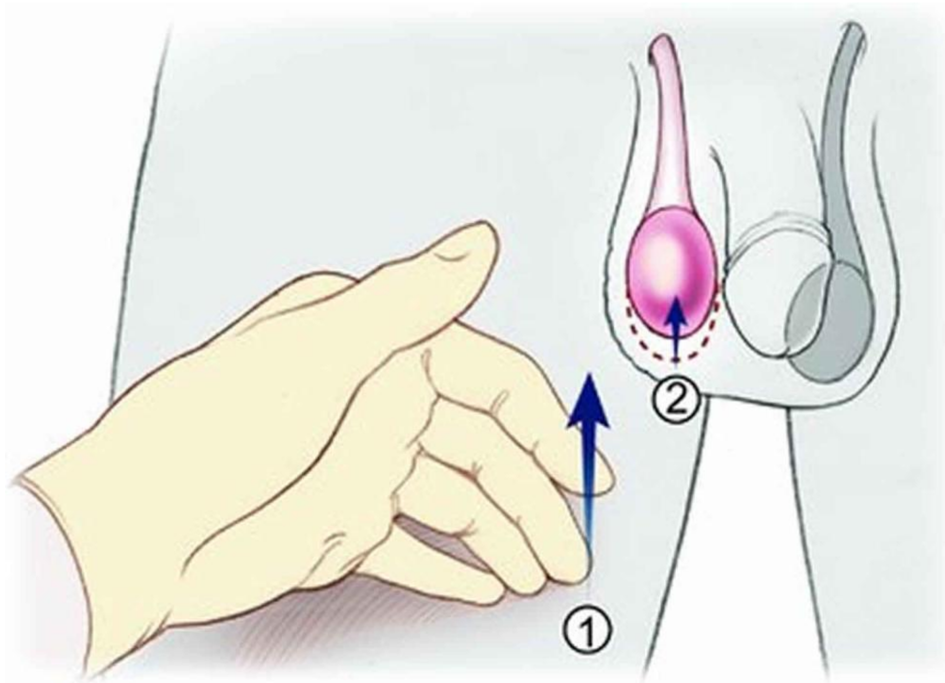
Cremasteric reflex

Stimulus	Afferent (sensory)	Integration center	Efferent (motor)	Effect
Lightly touching the skin over the medial side of the upper thigh.	Ilioinguinal nerve (L1)	Spinal cord at level L1	Genital branch of the genitofemoral nerve (L1,2)	Contraction of the cremaster muscle and elevation of the testis

* Strong in infants and adolescents



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Cremasteric reflex

- Ilioinguinal nerve supplies the upper medial portion of the thigh – this is the afferent limb of the cremasteric reflex arc
- Genital branch of the genitofemoral nerve supplies the cremaster muscle – this is the efferent limb of the cremasteric muscle reflex arc

Femoral branch of the genitofemoral nerve supplies the area over the femoral triangle

Lightly touching the skin over the medial side of the upper thigh invokes the cremasteric reflex – testis is pulled upward

Strong in infants and adolescents

Afferent pathway: ilioinguinal nerve *(L1)*

Efferent pathway: genital branch of the genitofemoral nerve *(L1,2)*

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Abdominal aorta and branches

- Continuation of the thoracic aorta
- Passes through the diaphragm at T12 vertebral level
- Gives paired and unpaired branches to both the abdominal viscera and abdominal wall
- Bifurcates at L4 into the common iliac arteries
- Common iliac arteries divide into internal and external iliac arteries
- Internal iliac artery supplies the pelvis and gluteal region
    - umbilical (partially obliterated), superior vesicle, middle rectal, **superior gluteal, inferior gluteal**, internal pudendal, **lateral sacral, obturator, iliolumbar**
- External iliac becomes the femoral artery as it passes deep to the inguinal ligament and supplies the lower limb

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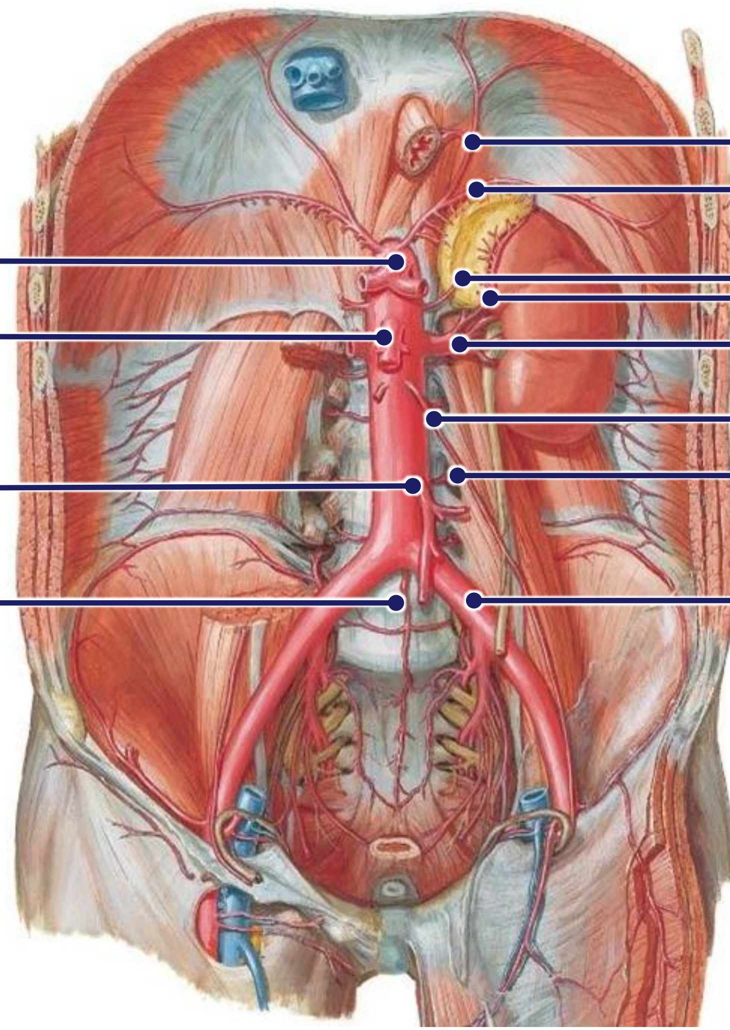

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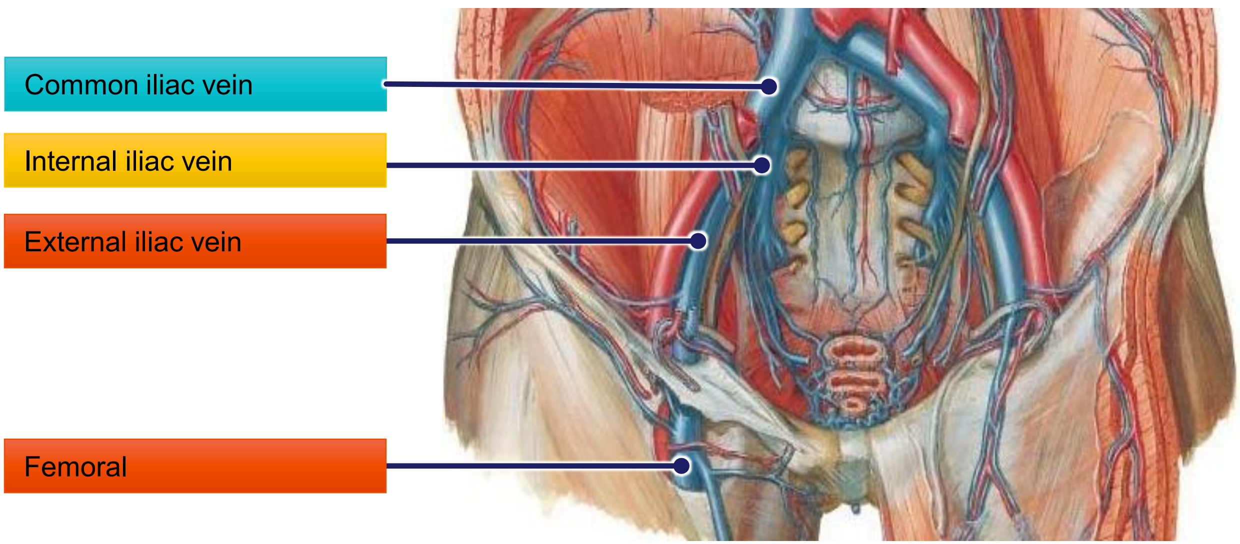
Common iliac veins

- Junction of the external and internal iliac veins at the pelvic inlet
- The internal iliac vein receives blood from the pelvic organs, pelvic wall, and gluteal region
- The external iliac vein receives blood from the lower anterior abdominal wall and lower limb



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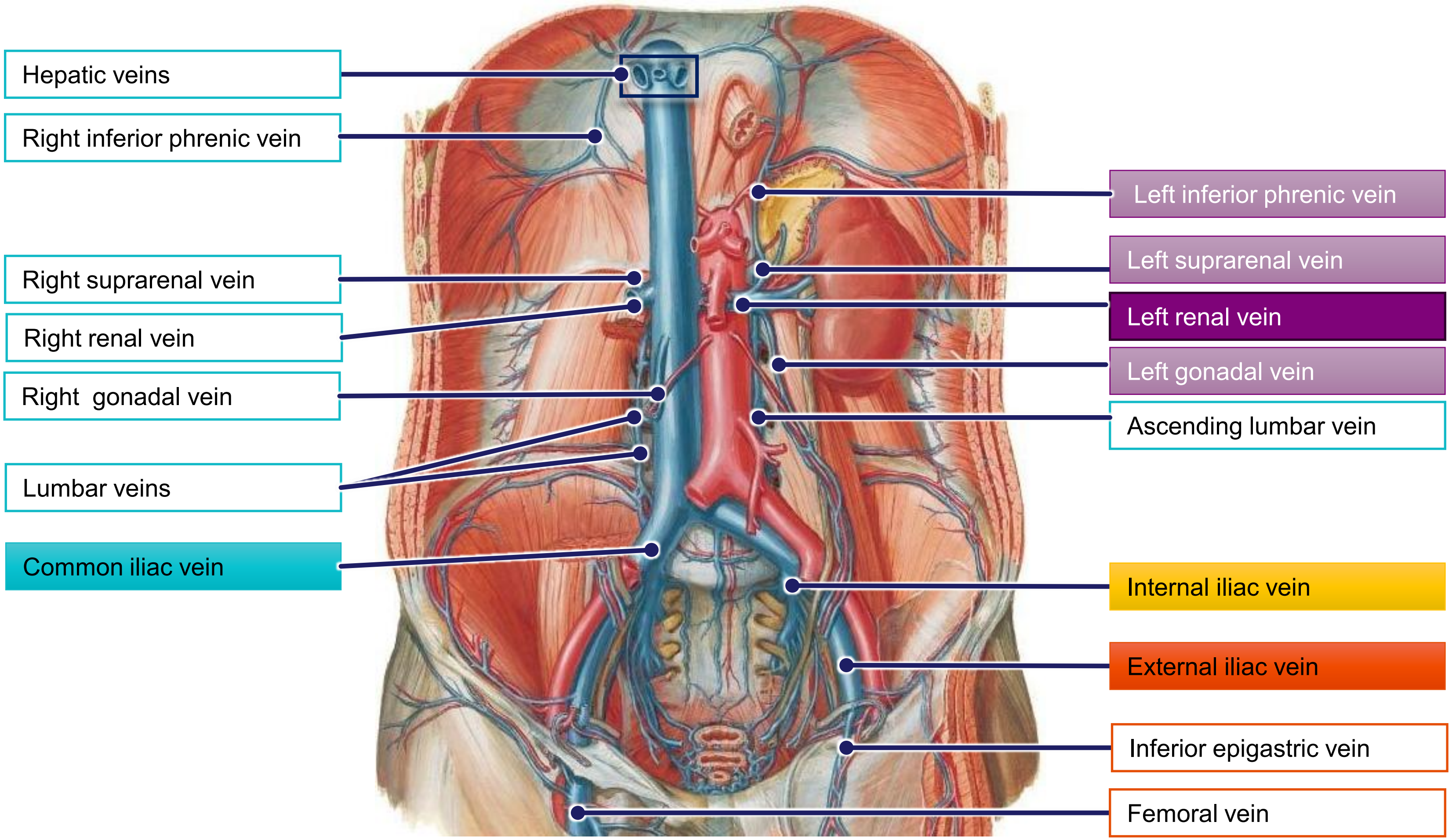

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Inferior vena cava

Starts at L5 vertebra with the junction of the two common iliac veins
The tributaries to the IVC do *not* correspond exactly to the branches of the aorta

It receives blood directly from the:

- Lumbar
- Right inferior phrenic
- Renal
- Right suprarenal
- Right gonadal
- Hepatic
- Common iliac veins

It receives blood indirectly from the:

- Left inferior phrenic (via renal veins)
- Left gonadal (via left renal)
- Left suprarenal (via left renal)
- Femoral and lower limb veins (via external iliac)
- Anterior abdominal wall (via external iliac)
- Pelvic organs (via internal iliac)

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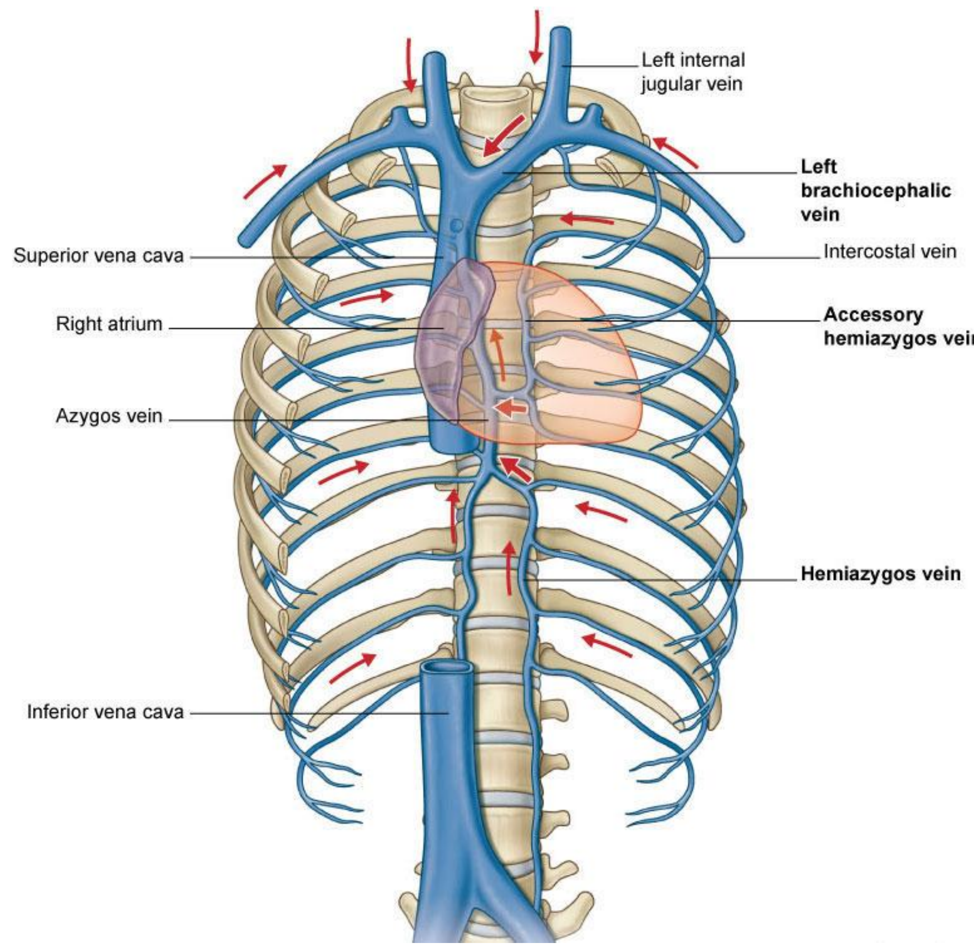
Azygos system of veins

Receives blood from the posterior walls of the thorax and abdomen into the superior vena cava

Formed by the union of the ascending lumbar veins with the right subcostal veins at the level of the 12th thoracic vertebra, ascending in the posterior mediastinum, and arching over the right main bronchus posteriorly at the root of the right lung to join the superior vena cava.

A major tributary is the hemiazygos vein, a similar structure on the opposite side of the vertebral column.

It also creates a **cavo-caval anastomosis** by offering an alternative, collateral blood flow from the lower half of the body to the superior vena cava, bypassing the inferior vena cava.



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Obstructed blood flow in the IVC

Azygos system of veins

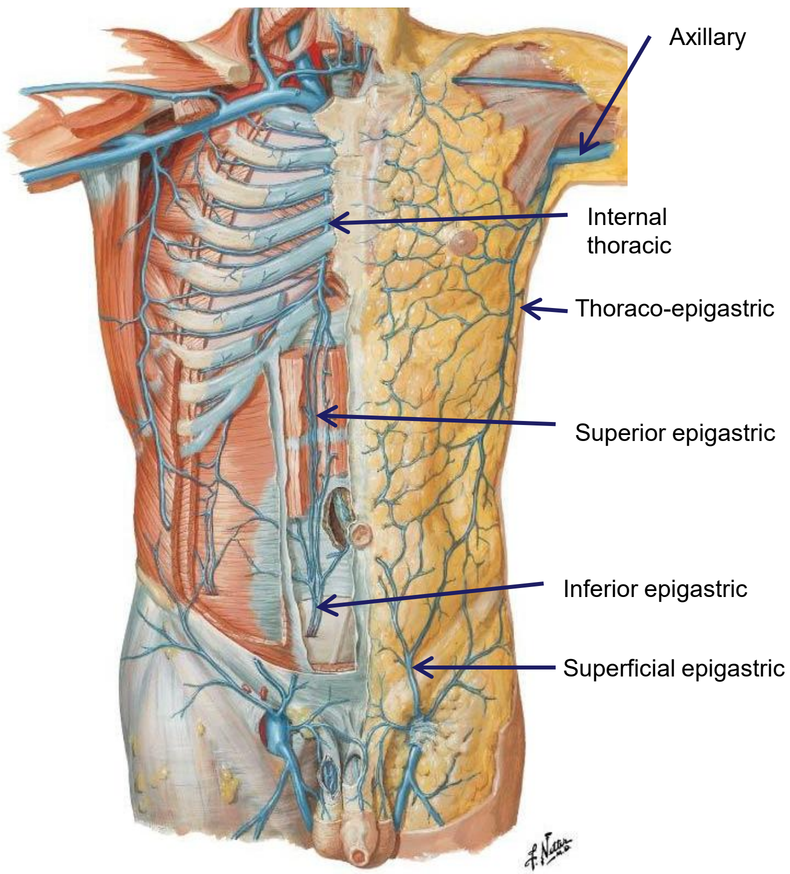
ascending lumbar veins receive blood from lumbar veins and drain into azygos or hemiazygos veins on to superior vena cava

Internal thoracic veins

continuous with superior and inferior epigastric veins and thus connects external iliac vein with subclavian vein on to right atrium eventually

Axillary vein

superficial epigastric veins anastomose with thoraco-epigastric veins which drain into the axillary vein which becomes subclavian, brachiocephalic to form superior vena cava with brachiocephalic vein from contralateral side



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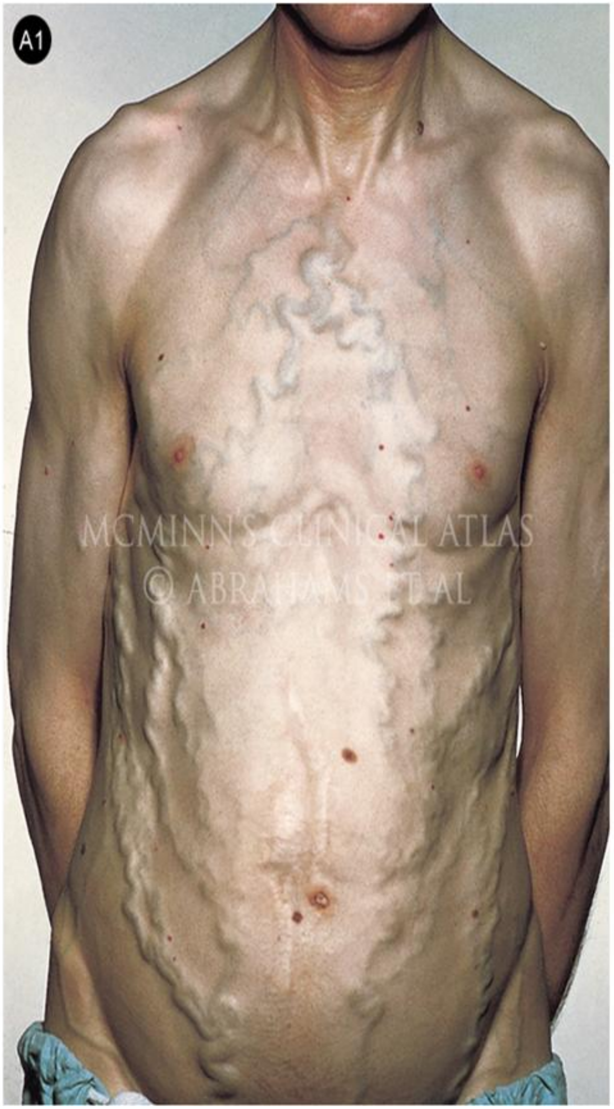
Obstructed blood flow in the IVC

Venous drainage from the abdominal and pelvic walls and pelvic organs is via the inferior vena cava

Pathologies can result in obstruction of the IVC

Several alternative drainage pathways are available

The alternative pathways may increase the size of the vessels. If these are superficial vessels, they may become visible under the skin



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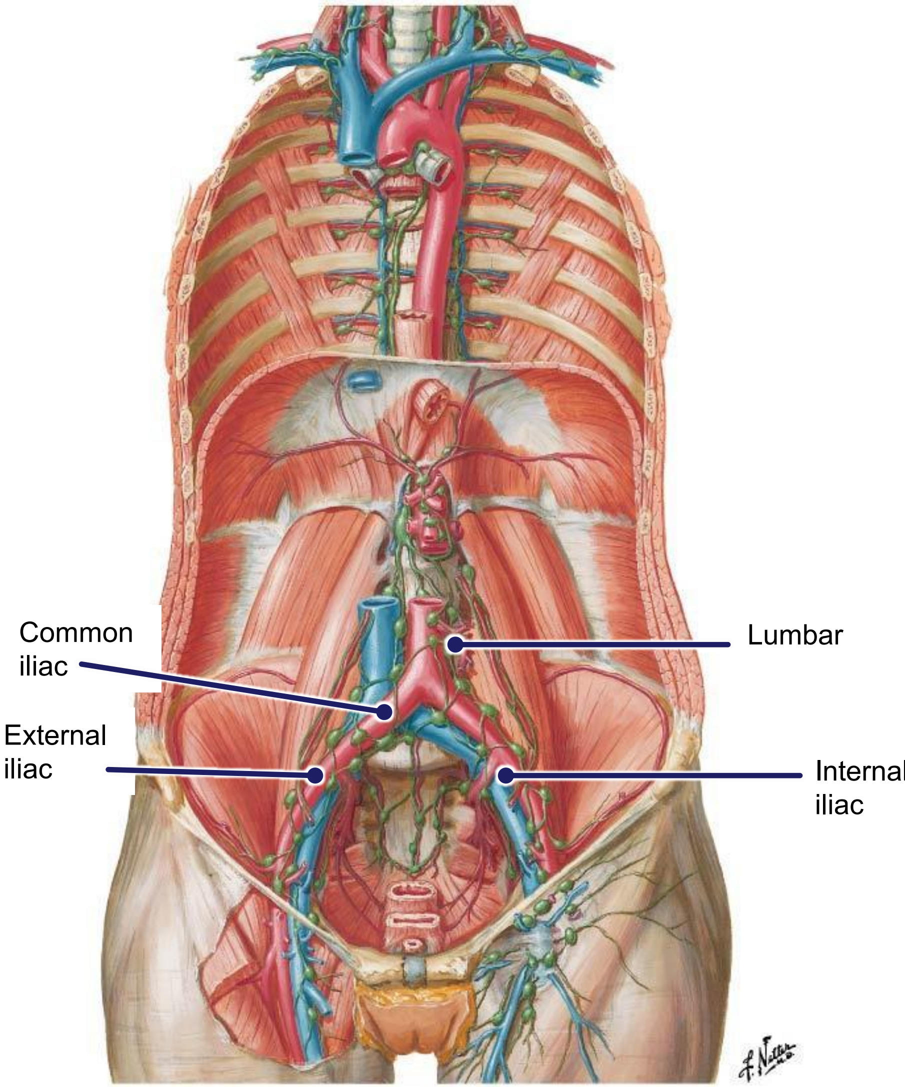
Abdominal & pelvic lymphatics



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Lymphatics from the posterior abdominal wall drain into the lumbar (lateral aortic) nodes

These eventually drain into the cisterna chyli, a large fluid-filled sac that collects lymph from the entire lower part of the body

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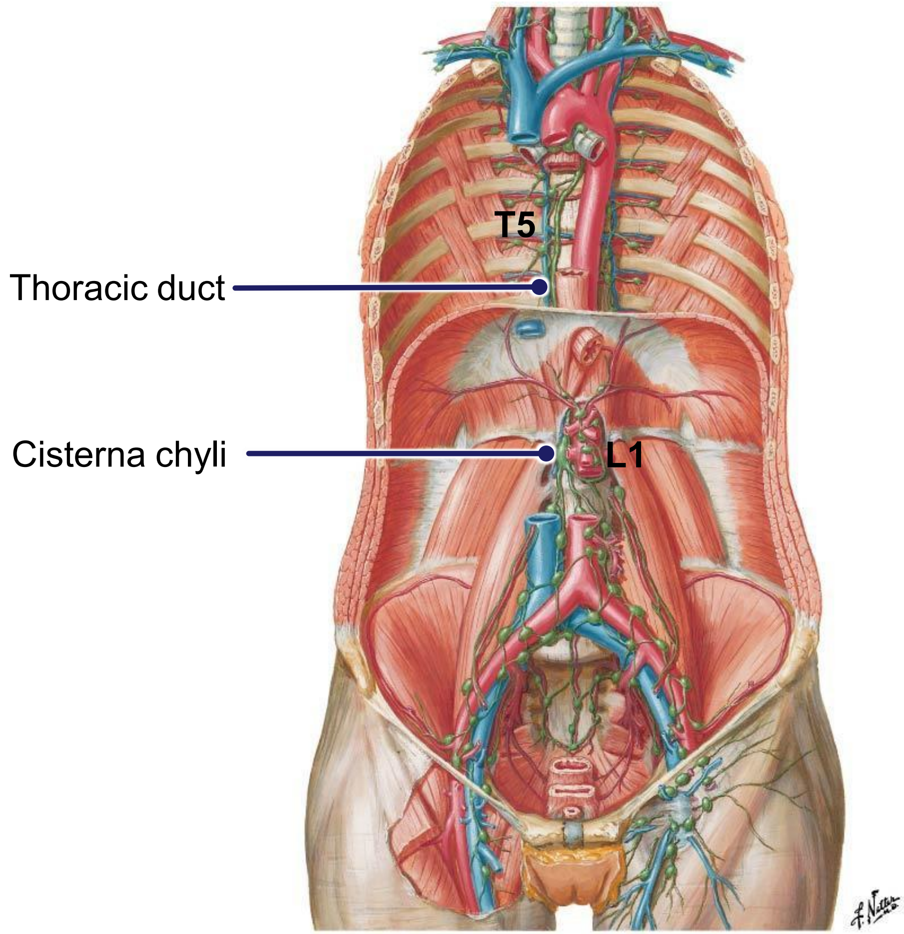
Abdominal lymphatics & cisterna chyli

Lymph from the posterior abdominal wall drains into the lumbar (lateral-para-aortic) lymph nodes which eventually drains into the cisterna chyli and continues as the thoracic duct

The cisterna chyli is a retro-peritoneal structure. In humans, it is located posterior to the abdominal aorta on the anterior aspect of the bodies of the first and second lumbar vertebrae (L1 and L2)

- it forms the beginning of the primary lymph vessel, the thoracic duct, which transports lymph and chyle from the abdomen cavity via the aortic opening of the diaphragm to the junction of left subclavian vein and left internal jugular vein

- It receives lymph from the intestinal trunk and the lumbar lymphatic trunks



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